Prototyping

- A prototype is a draft version of a product that allows you to explore your ideas and show the intention behind a feature or the overall design concept to users before investing time and money into development.
- A prototype can be anything from paper drawings (low-fidelity) to something that allows click-through of a few pieces of content to a fully functioning site (high-fidelity).

Benefits of Prototypes

- It is much cheaper to change a product early in the development process than to make change after you develop the site. Therefore, you should consider building prototypes early in the process.
- Prototyping allow you to gather feedback from users while you are still planning and designing your Web site.
- Nielsen has found that the biggest improvements in user experience come from gathering usability data as early as possible.
- He notes that it’s cheaper to make changes before any code has been written than to wait until after the implementation is complete.

High-Fidelity and Low-Fidelity Prototyping

- There is an on-going debate about using low versus high fidelity prototyping and how much a prototype should resemble the final version of your design.
- Both have been found to be basically equivalent in finding usability issues (Walker et al 2002). With that said, there are things to consider when trying to decide which option is best for your project:

High-Fidelity and Low-Fidelity Prototyping (Contd.)

- **Low-fidelity prototypes** are often paper-based and do not allow user interactions.
- They range from a series of hand-drawn mock-ups to printouts.
- In theory, low-fidelity sketches are quicker to create.
- Low-fidelity prototypes are helpful in enabling early visualization of alternative design solutions, which helps provoke innovation and improvement.
- An additional advantage to this approach is that when using rough sketches, users may feel more comfortable suggesting changes.

High-Fidelity and Low-Fidelity Prototyping (Contd.)

- **High-fidelity prototypes** are computer-based, and usually allow realistic (mouse-keyboard) user interactions.
- High-fidelity prototypes take you as close as possible to a true representation of the user interface.
- High-fidelity prototypes are assumed to be much more effective in collecting true human performance data (e.g., time to complete a task), and in demonstrating actual products to clients, management, and others.

The Difference between a sketch, a wireframe, and a prototype

- There are differences between a sketch, a wireframe, and a prototype, but how can we understand the distinctions and the best use of each?
- And what is their value as communication vehicles?
- Let’s discuss what separates a sketch from a wireframes from a prototype.

The Difference between a sketch, a wireframe, and a prototype (Contd.)

- **Sketches:**
  - Sketching and design emerged during the medieval period, so these concepts are nothing new. However, what I believe many people forget is that the intention of sketching is to separate design from the process of making.
  - In the context of design, *sketching* is rapid, freehand drawing that we do with *no* intention of its becoming a finished product.
  - **In fact, many times, sketching is only a foundation upon which we can overlay our actual design work.**
  - **Thus, sketching is a tool that supports the *process* of making, *not* the actual design itself.**
  - **Unfortunately, it's all too easy to forget this fact in our quest to get to an end product. We sometimes hold onto our sketches like they are the be-all and end-all of design.**

The Difference between a sketch, a wireframe, and a prototype (Contd.)

- **Sketches (Contd.):**
  - Sketches should be:
    - **quick**—Making them does not require long periods of time.
    - **timely**—You can create sketches on demand.
    - **inexpensive**—Creating them is cheap, using materials you have on hand.
    - **disposable**—You don’t care if sketches get thrown away.
    - **Plentiful** —Sketches don’t exist in isolation of one another.
    - **clear and use a common vocabulary**—Sketches comprise simple symbols and lines.
    - **fluid**—They have a fluidity of line and flow that imply creativity.
    - **minimally detailed**—Sketches are conceptual and suggest structure.
    - **appropriately refined**—They communicate just enough so others can understand your intent.
    - **suggestive and exploratory rather than confirming**—When sketching, you haven’t yet made any concrete decisions.
    - **ambiguous**—You have yet to work out the details, then overlay your design on the foundation your sketches provide.

The Difference between a sketch, a wireframe, and a prototype (Contd.)

- **Wireframes:**
  - A *wireframe*, in comparison, is a basic visual guide we use to suggest the structure of a Web site, as well as the relationships between its pages.
  - Wireframes come long before we do any visual design.
  - A wireframe's purpose is to communicate and explore the concepts that come out of sketching—that is, those concepts you actually want to pursue further during user interface design.
  - **Wireframing lets us outline a Web site's basic, overall structure and flow and helps us explore divergent ideas from our sketches.**
  - I like Will Evans' definition of wireframes in a recent article called "Shades of Grey: Wireframes as a Thinking Device." He says, "for me, wireframes act as a form of *thinking device* for the setting and exploration of a given problem space."
  - **You'll see there *isn't* much change from sketch to wireframe—merely a slight refinement and greater formalism.**
  - **Our initial drafts of wireframes remain tools or artifacts that we use during the process of making to aid conversations as design moves forward.**

The Difference between a sketch, a wireframe, and a prototype (Contd.)

- **Wireframes (Contd.):**
  - Wireframes should be:
    - **quick**—Making them does not require long periods of time, but they do take longer than sketches.
    - **inexpensive**—Creating them is cheap, using materials and tools you have on hand.
    - **viable**—Wireframes are *not* as plentiful as sketches, so you should have narrowed your concepts down to viable options.
    - **clear and use a common vocabulary**—Wireframes comprise simple symbols, lines, and indicators of interactivity.
    - **minimally detailed**—They are conceptual and suggest structure.
    - **appropriately refined**—They communicate a sense of structure and layout.
    - **confirmatory**—Wireframes present concepts that need validation. When creating wireframes, you still haven't made any concrete decisions.
    - **ambiguous**—You have yet to work out the finer points of interaction and content.

The Difference between a sketch, a wireframe, and a prototype (Contd.)

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The Difference between a sketch, a wireframe, and a prototype (Contd.)

- **Prototypes (Contd.):**
  - Prototypes should be:
    - **quick**—They should require only minimal development effort.
    - **inexpensive**—The development they require does not require a major investment.
    - **clear and use a common vocabulary**—Prototypes comprise simple symbols, lines, interactive components, and content.
    - **detailed**—Prototypes include content and interactivity.
    - **appropriately refined**—They represent an almost real application—though with a faked user experience.
    - **validated**—In prototypes, a design is no longer ambiguous or suggestive. A prototype is an actual instantiation of an idea. However, it may still require fine tuning.

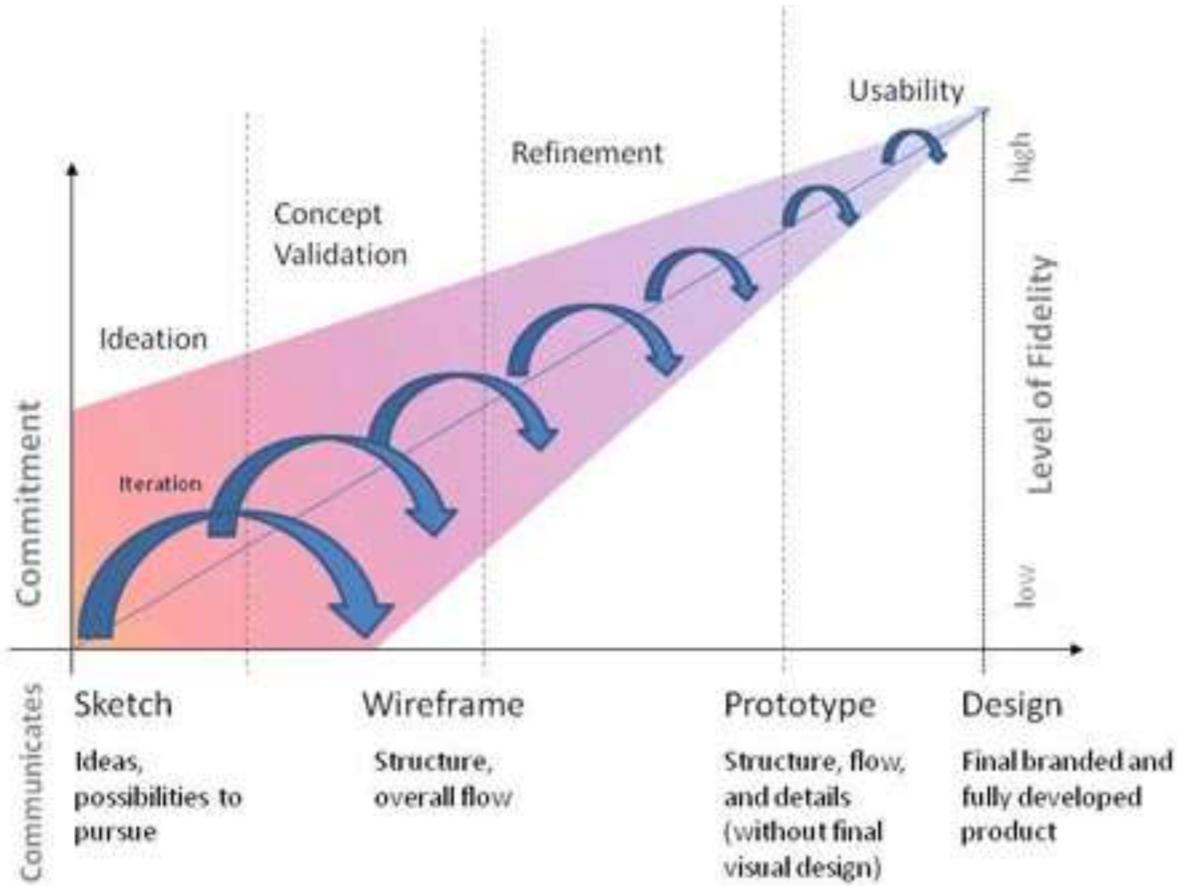
Understanding the Continuum

- Now comes the tricky part: deciding when a sketch stops being a sketch and becomes a wireframe, then a prototype.
- There is no clear boundary line separating these design artifacts, which is probably what makes our conversations about them so contentious and hard to articulate.
- Bill Buxton, in his book Sketching User Experiences, provides a list of descriptors that help explain the differences:

Sketch	Prototype
Evocative	Didactic
Suggest	Describe
Explore	Refine
Question	Answer
Propose	Test
Provoke	Resolve
Tentative	Specific
Non-committal	Depiction

Understanding the Continuum (Contd.)

- This illustration **depicts the relationships between a design's level of commitment and fidelity and what you are trying to communicate. These are the factors that matter when deciding where you are in this continuum. While the lines demarcating sketches, wireframes, and prototypes aren't clear, the phases of ideation, concept validation, and refinement and usability *are*—and that's where we should focus our attention.**



Understanding the Continuum (Contd.)

- **Sketches or wireframes support ideation and concept validation, while wireframes or prototypes are important for refinement and usability testing.**
- **When we are exploring and validating our divergent design ideas, a lack of commitment is important and beneficial. This allows us to be more flexible in our iterations and, therefore, more creative.**
- **As we start to whittle down our design options and refine those that merit further exploration, we need to start articulating them in a more realistic manner, so everyone on a product team can participate in the design conversation with the same understanding. This is why it's more important in later phases to create wireframes or prototypes.**
- There are also clear differences in the communication that happens during each phase that help distinguish which artifact is the best vehicle.
- And lastly, **you can see that the amount of iteration that occurs gets smaller and farther apart as we move toward the ultimate design. This is a natural progression as divergent ideas converge in an actual solution.**

References

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